MAKING MUSIC LEARNING FUN: DESIGNING AN INTERACTIVE IBOOK FOR INFORMAL LEARNING

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ABSTRACT

The use of new-format tablet computers (such as iPads) in educational settings has increased markedly since the first tablet computer was made commercially available. In secondary schools, the use of tablet computers has been used to help students learn a variety of subjects, particularly numeracy, literacy, and the sciences. The success of using tablet computers in schools is largely owing to the diversity of available electronic resources, such as Apps and web-based content that transform the boundaries between formal and informal learning. This is enhanced by the immediacy of access by students, both at school and at home. However, with the focus of interactive tablet content largely aimed at core subjects, there has been limited interactive content designed solely for music learning. In this paper we describe the design philosophy, and learning and potential teaching benefits of an interactive iBook that has been developed for secondary school students (years 7 and 8) to learn music. A demonstration of the iBook is also provided.

1. INTRODUCTION

Electronic books have been a disruptive force in educational innovation. The benefits of which have been assessed through worldwide pilot studies in a variety of subjects within different levels of education (Couse, Chen and Dora, 2010; Henderson and Yeow, 2012; Jennings et al., 2011; Kiger, Herro and Prunty, 2012). These pilots use different technologies, including the Kindle (Barack, 2011; Martinez-Estrada and Conaway, 2012) and the iPad (Olcese and Murray, 2011; Lau and Siu-lua, 2012). Owing to the design of the devices these technologies offer two different learning experiences: a passive experience similar to a paper book (the standard Kindle text-only experience), and an interactive experience that allows multi-touch, video and audio playback and user input (such as Apple's iBooks). The multifunctional aspect of the iPad, in particular, provides many options for both educators and learners to innovate and enhance their relevant experiences.

Many studies have examined the impact of mobile technology on pedagogy, and have found that not only do teaching practices change with technology (Bebell and Kay, 2010), but a large part of the successful adoption of technology in the classroom lies in the management of the technology (Henderson and Yeow, 2012), not just in terms of the practical management of the devices, but in the integration of the technology with pedagogy and content (Melhuish and Falloon, 2010; Mishra et al., 2011). For students, there is clear evidence that technology use in classrooms increases engagement, interest level, and achievement, as well as enhancing collaboration (Bebell et al., 2014; Bebell and O'Dwyer, 2010; Melhuish and Falloon, 2010). Students were also found to have increased motivation to learn independently (Kinash et al., 2012), which in the case of mobile devices that are accessible outside the classroom, is a critical part of informal learning. This research presents a set of clear design parameters when developing an interactive tool for learning music. This is shown in Figure 1.

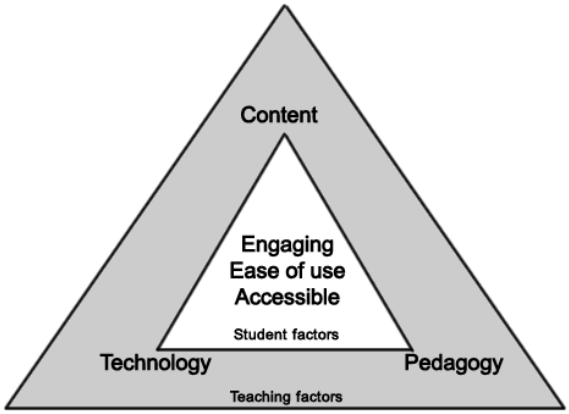


Figure 1. iBook design factors.

From a teaching perspective, the design of the iBook must be relevant in terms of content, technology, and pedagogy, in which these factors relate more to learning in a school environment. These factors should be considered in terms of appropriateness for the age group, curriculum, pedagogy and teaching philosophy, technological constraints and institutional policies that may promote or inhibit the use of such technology. Although teaching factors are a crucial component in the technology integration mix, there are student factors that relate to the success of the technology in both school, and out-of-school (informal) learning environments, with informal learning (in this case independent learning) being a critical component of a student's learning experience. A student's potential willingness or motivation to learn away from the classroom environment can be enhanced by addressing the student

factors and appealing to the student: by ensuring that the iBook and the content is engaging, easy to use, and easy to access outside of class.

This paper will focus on describing the design of an interactive music learning resource as an informal learning tool, from a student-centred design perspective. Although this approach omits teaching level factors, the synergy between student engagement and independent learning allows greater freedom to explore the interaction design opportunities on the iBook platform, for improving teaching and learning.

2. ADOPTED PEDAGOGY

Given the repeated themes of student motivation, independent learning and autonomy outlined in the literature around mobile learning, it was decided to adopt an established music pedagogy that also promoted these facets of learning.

Lucy Green's seminal text *How Popular Musicians Learn* (2002) included a series of case studies that explored that very topic, and summarised key learning features such as learning aurally by copying rather than from notation, learning in friendship groups, and learning music of their own choice. It established the term *informal learning* in music education.

In *Music, Informal Learning, and the School: a New Classroom Pedagogy*, Green (2008) extended this research with further case studies which transformed the initial findings into a pedagogical structure that could be used in schools. It was so successful that it was adopted as the model for the British intervention on music education called *Musical Futures* (Ainscough et al., 2009) which further consolidated Green's extensive research to a number of formulas that music educators could follow to bring informal learning through their music curriculum. Musical Futures was incredibly successful, with several studies showing that it improved student engagement, enhanced learning and enjoyment, and resulted in two to three times the number of students choosing to study music further (Hallam et al., 2009, 2011).

2.1. Modelling Aural Learning

The approach "Modelling Aural Learning" was adopted from Green (2008) and Ainscough et al.'s (2009) texts. In this variation on the informal learning pedagogy, students do not choose their own music, but the teacher selects music which is culturally relevant to students and which in their professional judgement they believe brings students great opportunity for musical learning.

Teachers are required to provide audio tracks of individual parts in the selected song, so that students do not have to pick one part out of a mastered song. Notation can also be provided, but it must be suitable for the genre and instrumentation provided: for example, guitar chord diagrams or tablature rather than traditional music notation.

Throughout development of the iBook content, the Modelling Aural Learning approach was followed.

3. DESIGN CONCEPTS

3.1. Interactive Elements

Apple's iBooks Author allows an iBook to exploit the multi-touch functionality of the iPad using widgets, to control a range of multimedia elements that can be embedded into the iBook. These multimedia elements include photo galleries, chapter reviews, audio, video, 3D objects, interactive images, "pop-overs", links and anchors, and html. Because "widgets" can be custom programmed using HTML, CSS and JavaScript, functionality can be extended. Such interactive elements have been utilised to great effect in educational iBooks such as E. O. Wilson's *Life on Earth* series (Wilson et al., 2014). In the *Life on Earth* series, the interactive elements assist the reader in understanding more complex scientific concepts, and are designed as sub-stories/explanations within the main theme of the book.

In contrast, using interactive elements to teach music performance and composition requires a different approach, revolving around the use of exemplars. For example, music performance can be taught using video performances of small excerpts, whereas video examples of some scientific concepts may prove too difficult. Similarly, an animation may be suitable to demonstrate cell division, but would be impractical to demonstrate fingering of long complex musical phrase on an instrument. In such a scenario, it is important to identify which interactive elements convey a concept in the most engaging and understandable way. Furthermore, the interactivity should be used in such a way as to be complimentary to the structure of the iBook.

3.2. Structure of the Book

The aim of this book is to teach students the fundamental elements of music composition and to develop their music performance skills by presenting new material for them to learn. In a sense, the iBook is intended to introduce a new structure for music learning, or a new context in which to apply skills previously learned. One such method for structuring such learning is by using "instructional scaffolding" (Bull et al., 1999). Bull et al. describe scaffolding in the context of technology in learning environments, and present concepts of scaffolding that can help not only organize the general structure of the book, but to organize the types of interactive elements that can be used to augment the various stages of learning. Bull et al describe these concepts as: advance organisers (the link between what is known and the new knowledge), modelling the appropriate behaviour), exploring new skills, and being generative with the learned information.

3.2.1. Advance organisers

One could argue that the iBook itself is an advance organizer, since it presents new information to learners. Conceptually, a positive way to motivate and engage students is to construct a scenario that can easily be related to. In the case of music, this may be situating the

book in context of an individual artist or genre of music. This information could be presented using audio, video and photos in order to make the 'iBook story' or theme more realistic.

3.2.2. Modelling

In real-world music learning, particularly instrumental learning, students often take part in one-to-one lessons with the teacher. In these lessons, the teachers will demonstrate the tasks (and/or behaviours) in a step-by-step manner that the students are expected to learn. Since the iBook will be a substitute teacher (during periods of informal, independent learning), it is important that these tasks continue to be demonstrated. One way that this can be done is through the use of video examples, either embedded in the iBook, or as hyperlinks to online video sites.

3.2.3. Exploring new skills

Teachers often encourage students to explore tasks with a view to the students accomplishing that task. In a musical context, one example might be for a student to explore the use of a scale or the exploration of a chord sequence. Exploring new skills using an iBook could include setting small challenges using 3D objects, animations, pop-overs, or even links to external online tests. It could also include getting students to play along with a video or audio excerpt, or by encouraging students to make recordings of their performances using third party apps.

3.2.4. Being generative with learned information

Students become generative with learned information when they finally have control. In the case of music, this could relate to learning a new chord sequence, and demonstrating to another learning, or by generating new contexts for the learned skill. This particular aspect is closely related to composition, since it aligns closely with understanding musical concepts and generating new music from those constructs (Hickey, 2003; Kaschub & Smith, 2013). In the context of an iBook, this could include an interactive graphic that asks students to re-order musical fragments, or to play a scale to music of a different genre. It could also include setting tasks that involve using third party apps to record or compose music. This work than then be presented in the school environment or collaborated on with peers.

4. DESIGN THEME: ELISHA KEEN

4.1. Developing a musical concept

The musical theme of the iBook is of paramount importance for student engagement. With significant number of choices in musical genre and musical context available to choose from, it is important to find a balance between student engagement, musical complexity and accessibility. The Modelling Aural Learning approach also dictates that music selected must be relevant to students' own musical cultures. Choosing a narrow theme with limited appeal, e.g. an unpopular genre of music, has the potential to limit student engagement with the learning resource (Hein, 2013). In addition, a genre of music that has a higher perceived complexity may also have similar detrimental effects. Both of these aspects of informal learning are related to the conceptual accessibility of the resource in terms of the link between what is known, and familiar, with the new knowledge. In other words, students should be at least familiar with the genre. The accessibility of the resource for the student extends beyond knowledge linking and advance organization, into a priori cultural, personal and individual experience, and includes gender neutrality, and aspirational motivation. Consequently, the music of singer/songwriter/performer Elisha Keen was chosen as a case study for the iBook. In order minimize the complexity of the iBook, the resource examines her song "Breathe Some Different Air". This song is a contemporary ballad that uses a typical popular song structure (verse/chorus/verse etc.).

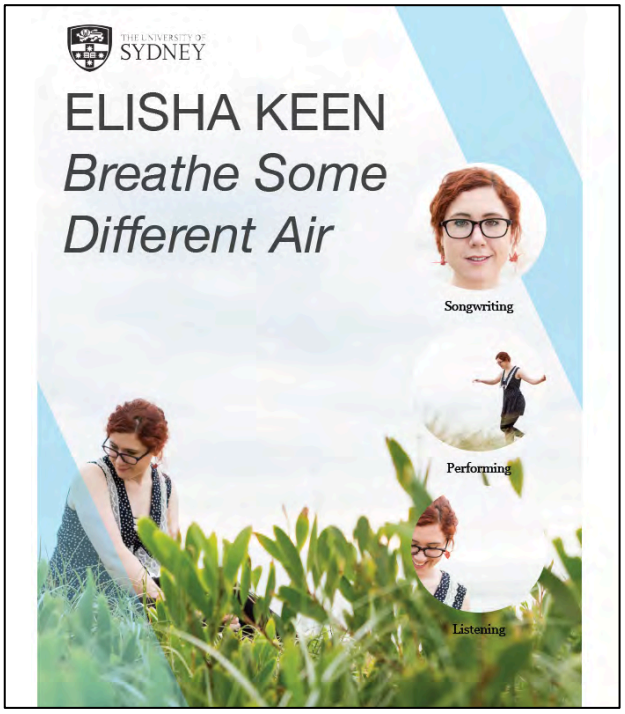


Figure 2. The front page of the iBook.

To increase Elisha Keen's accessibility with students, a video of Elisha talking about her songwriting processes are embedded into the early pages of the iBook. This has a dual purpose. First, Elisha discussing her songwriting processes is presenting new information to students. It seeks to motivate and encourage students to continue. Second, the video provides a modelling exemplar, whereby the process of composing a song is demonstrated to students. Furthermore, the interview was filmed in the recording studio to add aspirational qualities to the narrative.



Figure 3. An iBook page with an embedded video interview with Elisha Keen.

4.2. Modelling and Exploring using video and audio

In Modelling Aural Learning, in the absence of a teacher providing examples in person and teaching didactically, it is still possible to provide a mechanism for demonstrating, or modelling, the tasks or skills. Since users of the resource may have different skill levels, and may not be familiar with the song to be played, it is important that the modelling continues the step-by-step demonstrations. In this iBook, a layered approach was taken, where modelling was designed to incrementally add examples that relate to different levels of compositional complexity. The interactive element chosen for the modelling of the first layer, is video of how to play the chord progressions on the guitar. This video is accompanied by tablature outline the chord progressions. One of the benefits of using this resource lies in the ability for the learner to play back the video until they have learnt the sequence.

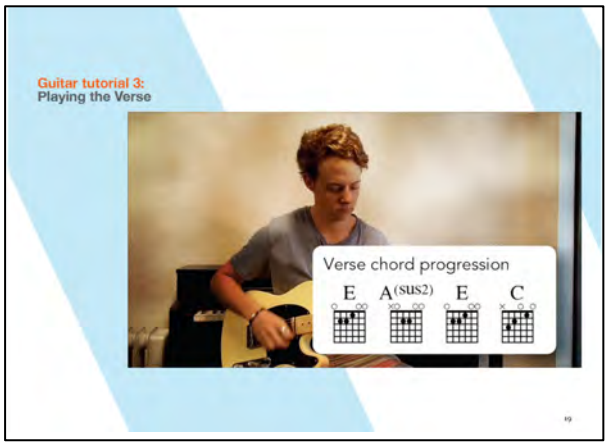


Figure 4. Modelling of the instrumental performance is done using video and graphics for the individual sections of the song.

In order to assist in the modelling process, students are given exemplars in the form of audio files of individual parts that demonstrate the different sequences. Elisha Keen provided the original stem recordings, so that examples are the same high standard as the final mixed song. This gives students the opportunity to explore their knowledge of the individual chords in the context of longer sequences of chords and with different instrumental contexts, or musical layers. This modelling approach is continued and expanded upon to include different instruments, including the vocal tracks, bass guitar and drums, with minimal notation.

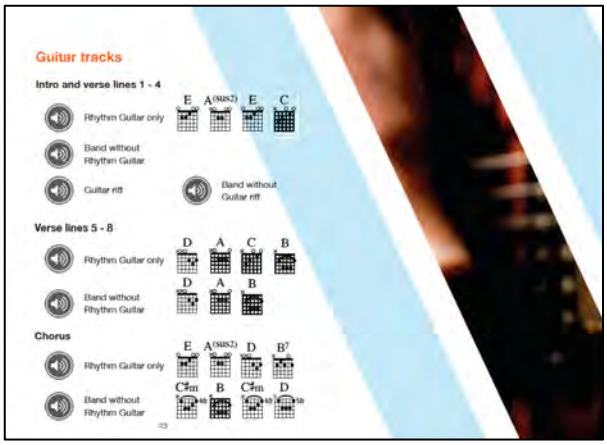


Figure 5. Students are given exemplars of the guitar sections in the form of audio examples, and guitar tablature for them to play along.

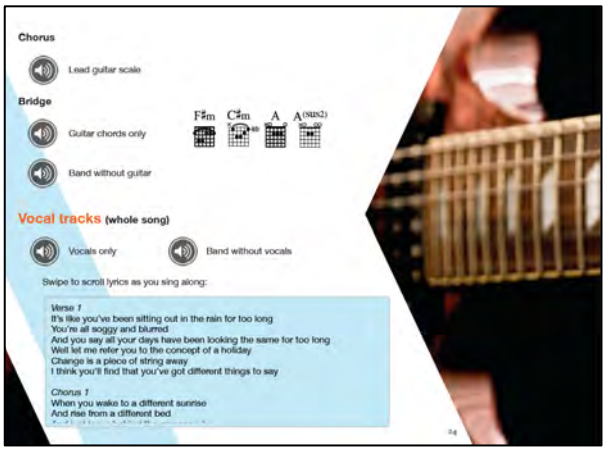


Figure 6. Students are also given vocal track exemplars.

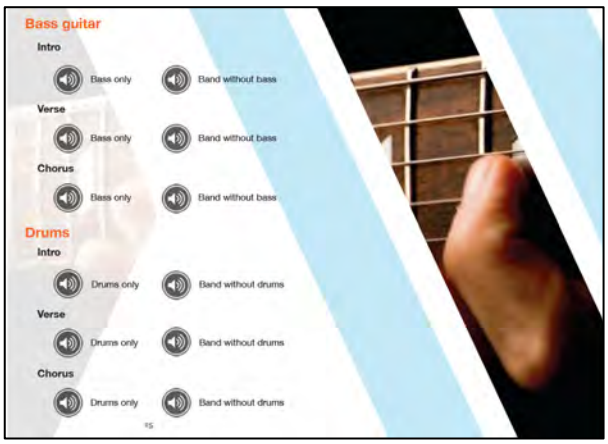


Figure 7. Students are given bass guitar and drum exemplars.

By adding the exemplars of additional instruments, students are given the opportunity to hear different elements of the song in isolation. It also increases the accessibility of the song as they are exposed to the constituent components without interference, in order to assist in the development of additional skills, such as listening. Furthermore, this format encourages the student to explore newly learned skills by giving students the opportunity to play along with individual instruments, even though this may not be expressly stated in the iBook. Once the different sequences of chords have been learned, these sequences are contextualized in chord sheet that matches the chord sequences with the lyrics. This adds a final layer of complexity to the learning of this song.

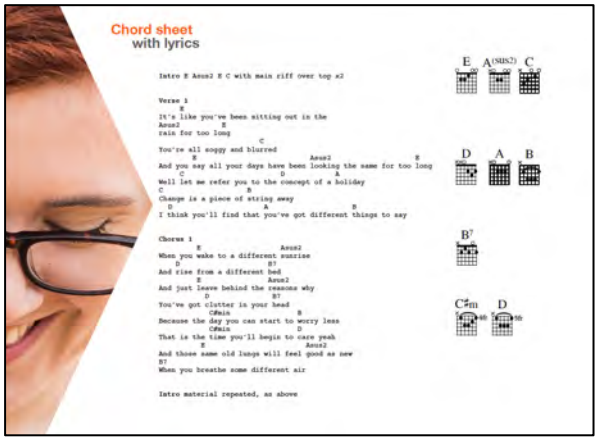


Figure 8. Students are given a chord sheet that relates the chord sequences to the vocals.

4.3. Being generative using interactive animation

The final part of the iBook enables students to be generative with the learned information, specifically in relation to the song structure. In order to allow students to expand create new sequences, an interactive animation allows the student to select the different parts of the song, reorder them, and listen back to it.

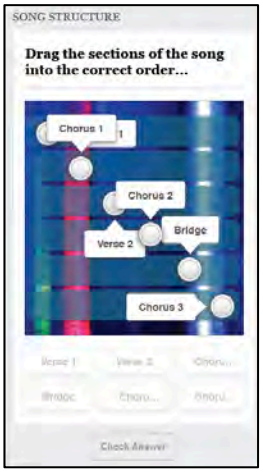


Figure 9. A task for students to explore new information. Students are asked to re-order sections of the music to see what the effects are.

This activity was designed as an exploratory tool to enable students to gain an understanding of song structure, and to create and listen to new structures. Furthermore, this activity tests the students' knowledge and engages students in thinking about the song from a structural and holistic perspective.

5. CONCLUSIONS

As a starting point, the iBook content being informed by key aspects of informal learning pedagogies creates synergies with the framework for instructional scaffolding, whereby many factors that that are designed to engage, encourage and increase accessibility of the content are inherent in the interactive objects. For example, the video tutorials with graphics on how to play a chord sequence, and the interactive animations for re-ordering song sections both inherently engage, encourage and make the content accessible in a manner that is familiar to music learning.

By drawing upon concepts of instructional scaffolding, the iBook clearly presents a narrative. Content that serves as advance organisers are augmented by interactive elements to ensure clarity, and relevance to informal and independent learning. The exemplars that demonstrate how to play various parts of the song not only model appropriate behaviour and skill, but as a collection of exemplars across different instruments allow greater levels of conceptual complexity, while simultaneously allowing students the opportunity to explore new skills in different contexts.

The separation of musical elements from the song also allows students with different instruments to use the resource, and to enhance their learning. Furthermore, this approach enables students learning different instruments to collaborate musically by either performing together, composing together, or simply by teaching other students. Finally, the interactive animation is the most appropriate method for encouraging students to be generative with learned information.

One conclusion that can be drawn from the iBook design framework, is that there are at the very least some basic principles in interactive iBook design that should always be considered. This relates specifically to the type of interactive element used to convey a particular teaching element. In this paper, videos have been used in predominantly the iBook narrative and instructional modelling. In these two areas of the iBook, students are "learning". In other areas, such as exploring new skills and being generative with learned information, students are "participating" (even though they are still learning). In contrast, audio is predominant throughout the iBook, and as a ubiquitous element, transcends instructional boundaries. Thus, it is worth considering the objective outcome of each section within the context of the learning experience, as to the type of interactive object used.

6. FURTHER WORK

Further work is underway to develop the iBook resource to include composition frameworks, testing, and teaching factors, particularly in relation to curricular alignment, in order to evaluate the resource as a formal learning tool. In addition, further work could include trialling the iBook in schools to assess the impact of such electronic resources on teachers and students in music education. This could include extending the iBook content to different themes, including different genres of music, or including music from different cultures.

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